

Introductory Statistics

2025 Lectures Part 2 - Data and Statistics

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Statistics around us

Examples of reports in media:

- With respect to last week, the average price of the Natural 95 petrol decreased by 1 heller to CZK 36.47 a liter in the Czech Republic. Further development of the price is uncertain. (ČTK České noviny, February 13, 2025).
- The percentage of the economically active persons aged 15-64 years reached 77.9 % in December 2024 and it increased by 0.8 percentage point compared to that in December 2023. (ČSÚ, January 31, 2025).
- In Q3 2024 the median wage (CZK 40 482) increased by 6.6% compared to the corresponding period of the previous year. (ČSÚ, December 4, 2024).
- The Dow Jones Industrial Average closed at 44,546.08 (The Wall Street Journal, February 14, 2025).

Descriptive statistics refers to numerical facts such as averages, medians, percents, and index numbers that help us understand the reality via available data.



SHOULD I HAVE THE SOUP?
EVERYTHING ELSE IS SO
EXPENSIVE, AND I DON'T
KNOW WHO'S PAYING... ARE
STATISTICIANS **STINGY**? I'VE
NEVER GONE OUT WITH
ONE BEFORE... THOUGH I
ONCE KNEW A **VERY**
GENEROUS ACCOUNTANT...

SHOULD I HAVE THE SOUP?
27 OUT OF THE **36** TIMES
I'VE HAD IT, IT WAS PRETTY
GOOD... BUT IS MONDAY THE
REGULAR CHEF'S NIGHT
OFF? AND WHAT IF ALL THE
AIR MOLECULES IN THE
ROOM SUDDENLY FLY UP TO
THE CEILING?



Subject of Statistics

- in most scientific/industrial studies decisions are made based on **data** - past experiences/observations, results of some controlled process/experiments
- producing data enables analysis and drawing useful conclusions from them
- an inherent part of decision making is the **knowledge of relevant information/data** to the decision (possibly unknown or subject to future events/uncertainty) - to reduce guesswork
- **statistics as a science**: to analyze/approximate a part of reality based on limited information – **science of data**
- statistics involve collecting, classifying, summarizing, organizing, analyzing, presenting and interpreting data and also model building
- application in agriculture, astronomy, biology, business, economics, education, electronics, geology, medicine, engineering, weather forecast etc.



Application in Economics and Finance

- **accounting** - sampling procedures when conducting audits
- reviewing and validating every account can be too time-consuming and expensive - select and review a subsets of accounts and draw a conclusion about all accounts
- **finance** - statistical information as a guide to investment recommendations - e.g. in case of stocks review financial data including price/earnings ratios and dividend yields to draw a conclusion whether a stock is over- or underpriced
- **marketing** - electronic scanners and retail checkout counters collect data for a variety of marketing research applications - brand managers can review the scanner statistics to establish future promotional activities
- **economics** - statistical data for variety of forecasts about the future of aspects of economy - e.g. forecasting inflation rates, economists enter various indicators (Producer Price Index, unemployment rate, etc.) into a forecasting model

- **data** are facts and figures collected, analysed and summarized for presentation and interpretation
- **data set** is a collection of all data in a particular study
- **elements** are the entities on which data are collected
- **variable** is a characteristic of interest for the elements
- **measurements** on each variable for every element in the study provide data set
- **observation** is the set of measurements for a particular element



Example Morningstar

Example 1: Consider the following data set containing information for 25 mutual funds that are part of the Morningstar Funds 500 for 2008.

Fund Name	Fund Type	Net Asset Value (\$)	5-Year Average Return (%)	Expense Ratio (%)	Morningstar Rank
American Century Intl. Disc	IE	14.37	30.53	1.41	3-Star
American Century Tax-Free Bond	FI	10.73	3.34	0.49	4-Star
American Century Ultra	DE	24.94	10.88	0.99	3-Star
Artisan Small Cap	DE	16.92	15.67	1.18	3-Star
Brown Cap Small	DE	35.73	15.85	1.20	4-Star
DFA U.S. Micro Cap	DE	13.47	17.23	0.53	3-Star
Fidelity Contrafund	DE	73.11	17.99	0.89	5-Star
Fidelity Overseas	IE	48.39	23.46	0.90	4-Star
Fidelity Sel Electronics	DE	45.60	13.50	0.89	3-Star
Fidelity Sh-Term Bond	FI	8.60	2.76	0.45	3-Star
Gabelli Asset AAA	DE	49.81	16.70	1.36	4-Star
Kalmar Gr Val Sm Cp	DE	15.30	15.31	1.32	3-Star
Marsico 21st Century	DE	17.44	15.16	1.31	5-Star
Mathews Pacific Tiger	IE	27.86	32.70	1.16	3-Star
Oakmark I	DE	40.37	9.51	1.05	2-Star
PIMCO Emerg Mkts Bd D	FI	10.68	13.57	1.25	3-Star
RS Value A	DE	26.27	23.68	1.36	4-Star
T. Rowe Price Latin Am.	IE	53.89	51.10	1.24	4-Star
T. Rowe Price Mid Val	DE	22.46	16.91	0.80	4-Star
Thornburg Value A	DE	37.53	15.46	1.27	4-Star
USAA Income	FI	12.10	4.31	0.62	3-Star
Vanguard Equity-Inc	DE	24.42	13.41	0.29	4-Star
Vanguard Sht-Tm TE	FI	15.68	2.37	0.16	3-Star
Vanguard Sm Cp Idx	DE	32.58	17.01	0.23	3-Star
Wasatch Sm Cp Growth	DE	35.41	13.98	1.19	4-Star

Source: Morningstar Funds 500 (2008).



Scales of measurements

- **nominal scale** - data for a variable consist of labels or names used to identify an attribute of the element. For numerical purposes may be replaced by a numerical code 1, 2, E.g. Fund type, sex, nationality.
- **ordinal scale** - data exhibit the properties of nominal data and the order or rank of the data is meaningful. Ordinal data can also be provided using a numerical code. E.g. Morningstar Rank, highest achieved education
- **interval scale** - data have all the properties of ordinal data and the interval between values is expressed in terms of a fixed unit of measure. Interval data are always numerical. E.g. year of birth, temperature in degrees Celsius.
- **ratio scale** - data have all the properties of interval data and the ratio of two values is meaningful. This scale requires that zero value be included. The scale of most of the variables we measure is a ratio scale. E.g. cost of a car, time, weight, distance.



Another types of classification

Categorical vs Quantitative:

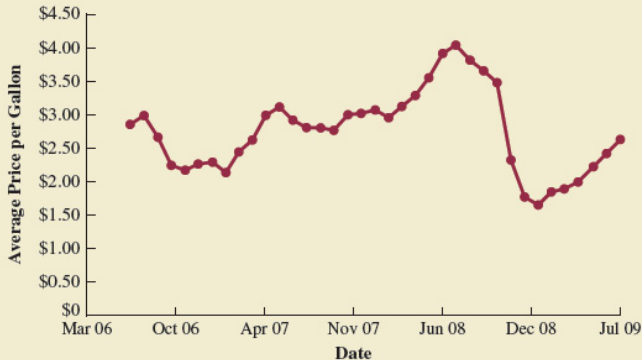
- **categorical data** - data grouped by specific categories; of either nominal or ordinal scale of measurement. Sometimes referred to as **qualitative data**. Statistical analysis of categorical variables is limited, we can summarize by counting number or proportion of observations in categories. It may not have sense to consider characteristics such as sum or average.
- **quantitative data** - data with numerical values indicating “how much” or “how many”. Arithmetic operations provide meaningful results for quantitative variables.

Cross-sectional vs Time series

- **cross-sectional data** - collected at the same or approximately the same point of time. E.g. Morningstar data set
- **time series data** - collected over several time periods. E.g. average price per gallon of gasoline between 2006 to 2009.

Example Average price per gallon of gasoline

Example 2:



Source: Energy Information Administration, U.S. Department of Energy, July 2009.



Existing data sources

- data can be obtained from existing sources or from surveys and experimental studies designed to collect new data
- there are organizations that specialize in collecting and maintaining data, e.g. Bloomberg, Dow Jones & Company, Median, SCIO, Ministry of Finance, Czech Statistical office, Google
- examples of data from internal company records:
 - employee records: name, address, salary, number of vacation days, number of sick days, bonus
 - production records: part or product number, quantity produced, direct labor cost, materials cost
 - sales records: product number, sales volume, sales volume by region or customer type
 - credit records: customer name, address, phone number, credit limit, accounts receivable balance





Macroeconomic Forecast - January 2025

The Macroeconomic Forecast is prepared on a quarterly basis. It contains a forecast for the current and following years and for certain indicators an outlook for another 2 years.

07.03.2025



We Published

Minister

Results of T-Bonds Auctions - 2025

13. February 2025

Results of T-Bills Auctions - 2025

13. February 2025

Announcement on State Treasury Bills Auction (3rd Tranche of 922nd Issue)

13. February 2025



Zbyněk Stanjura

MINISTER'S MEDALLION

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30/09/2024

10,897,237

Publication date:
30. 12. 2024



[Average gross wages](#)



2024, quarter 3

CZK 45,412

Publication date:
04. 12. 2024



[Gross domestic product](#)



2024, quarter 4

1.6%

Publication date:
31. 01. 2025



[Inflation rate](#)



2024

2.4%

Publication date:
13. 01. 2025



[Other data](#)

News

17
February
2025

[Employment and unemployment as measured by the LFS - 4. quarter of 2024](#)

[Analysis](#) [Employment and unemployment \(LFS\)](#)

12
February
2025

[Consumer price indices - inflation - January 2025](#)

[News Release](#) [Prices, inflation](#) [Inflation, consumer prices](#)

Statistical studies

- sometimes data are not available - can be obtained by conducting a statistical study
- in an **experimental study** a variable of interest is first identified. Then one or more other variables are identified and controlled so that data can be obtained about how they influence the variable of interest. E.g. effects of a new drug on blood pressure.
- **observational or non-experimental studies** make no attempt to control the variables of interest. E.g. survey, when research questions are first identified and a questionnaire is designed and administered to a sample of individuals
- the type of source used depends on time and cost required to obtain the data



Descriptive statistics

- most of information in newspapers, company reports, etc. consists of data summarized in a form that is easy for the reader to understand (tables, graphs, numerical values) - **descriptive statistics**. They refer to a specific data set, results cannot be generalized to full population.

Example 1 (cont.):

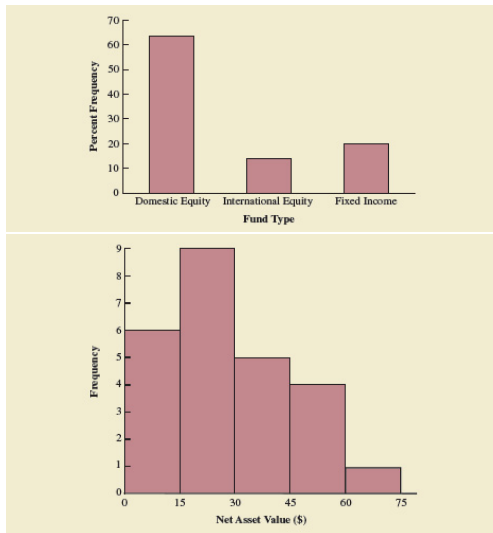
Tabular summary:

Mutual Fund Type	Frequency	Percent Frequency
Domestic Equity	16	64
International Equity	4	16
Fixed Income	5	20
Totals	<u>25</u>	<u>100</u>



Descriptive statistics

Graphical summary:



Statistical inference

- we often seek information about large group of elements - **population** - set of all elements of interest in a particular study (e.g. voters, companies, products etc.) but because of some reason (time, cost, etc.) data are collected from only a small portion of a group - **sample** - a subset of population
- conducting a survey to collect data for the entire population is called a **census**; collecting data for a sample is called **sample survey**
- statistics uses data from a sample to make estimates or test hypotheses about the characteristics of a population through a **statistical inference**



